AI for Climate

Meiyu Prepare

Rainfall forecasts and stock relocation decisions for small shops

A working prototype for comparing expected costs

Built by two students. Merchant validation is pending.

The small shop dilemma

Target users: small shops in Hangzhou preparing for wet weather

Relocate now

Pay for labour and materials.
Reduce stock loss if an event occurs.

Keep stock in place

Avoid the immediate relocation cost.
Leave the stock exposed.

Weather is uncertain. Preparation has a cost.
The prototype makes the assumptions and calculation visible.

A single 24-hour decision

Choose a weather window, then enter your own cost assumptions

01

Weather window

Location and 24-hour window
Capture time
Data completeness

02

Cost assumptions

Relocation cost C
Event loss if kept L_0
Event loss after relocation L_1

03

Cost comparison

Expected cost comparison
Sources and assumptions

Missing members, missing hours or stale data stop the weather calculation.

A paired dataset with 24-hour windows

Fixed-lead ECMWF forecasts paired with Xiaoshan Airport SSOD reference data

Window (UTC+08)	Forecast (mm)	Reference (mm)	Quality
2025-06-01 08:00 to 2025-06-02 08:00	52.5	29.0	passed

731

Calendar days

670

Complete reference days

636

Complete paired days

Exclude windows with a required hour missing, invalid data or negative rain.
Never fill missing observations with zero.

Historical model and online ensemble

Historical offline study

52.5 mm

Forecast for 1 Jun 2025

log1p and standardization
Logistic regression

67.9%

P(reference daily rainfall ≥ 25 mm)

Train 276 days, validation 120 days
Test 240 days with 18 events

Archived online snapshot

51 members × 24 hours

20 Oct 08:00 to 21 Oct 08:00 (UTC+08)

Sum rain for each member
Count totals at or above 25 mm

0 / 51 = 0%

Uncalibrated ensemble member fraction

Captured 19 Oct 2026, 19:22 (UTC+08)

Neither output represents a shop flood or stock loss probability. The historical model serves offline research only.

A CNY 460 gap in expected cost

Hypothetical inputs, separate from weather forecasts

Event losses: L_0 = CNY 1,000 if kept, L_1 = CNY 100 after relocation

Relocation cost C = CNY 80 (labour 60 + materials 20)

Assumed event probability p = 60%

Keep stock in place

$pL_0 = 0.60 \times 1{,}000$

CNY 600

Relocate

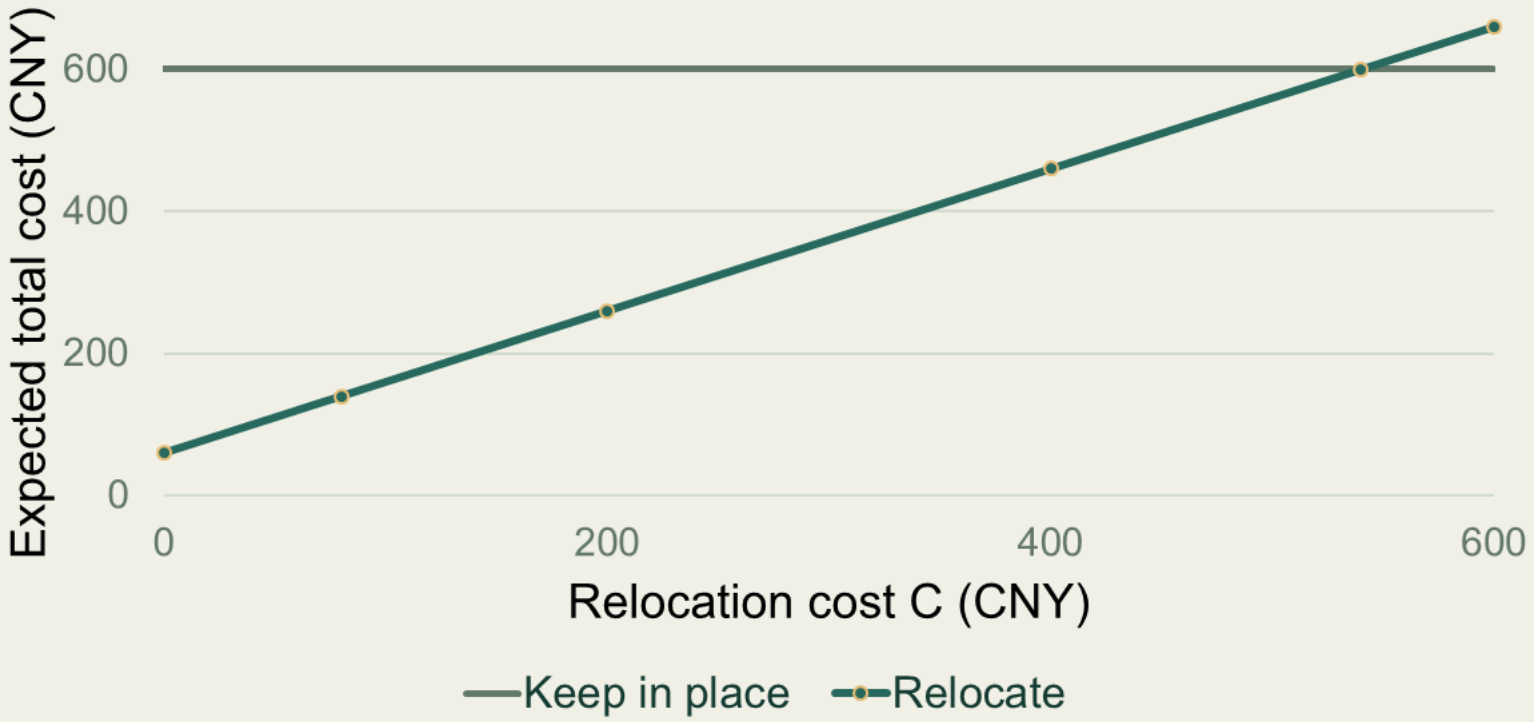
$C + pL_1 = 80 + 0.60 \times 100$

CNY 140

The CNY 460 difference comes from this scenario. Actual merchant loss reduction is unverified.

Relocation cost changes the choice

Same assumptions: p = 60%, L_0 = CNY 1,000, L_1 = CNY 100



C = CNY 80

Relocate: CNY 140
Keep: CNY 600

C = CNY 540
Equal expected costs

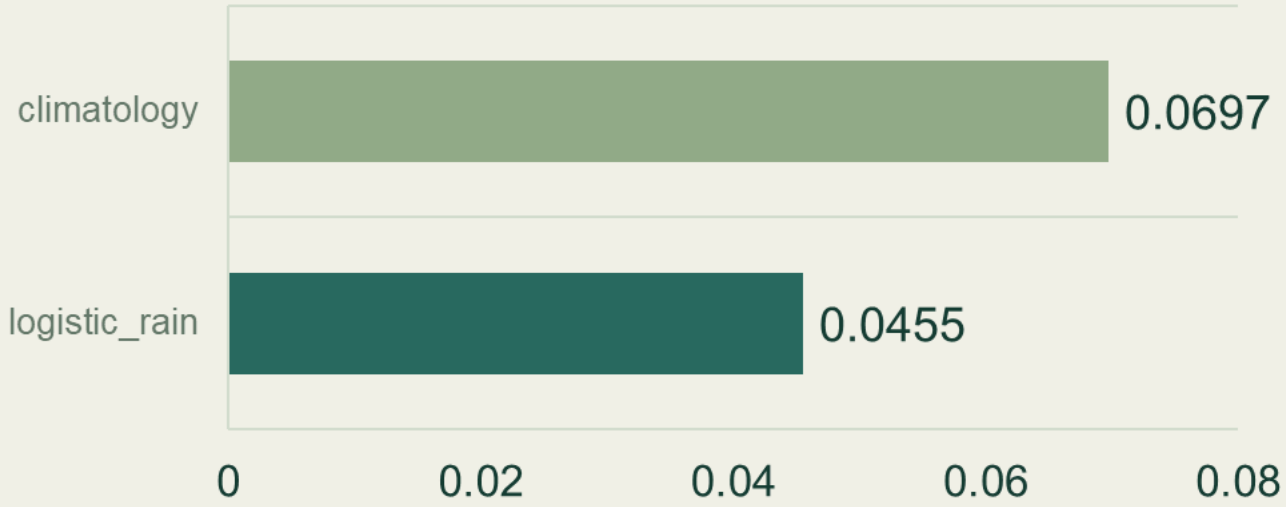
C = CNY 600
Keep stock in place

Break-even cost C* = $p(L_0 - L_1)$ = CNY 540. Inputs determine the comparison.

Evidence and limits

Offline study: the same 240 test days with 18 reference events

Test Brier score (lower is better)



Climatology uses the training event rate.

Only 18 test events
Reference labels have limits

Online fraction still needs local calibration

Merchant loss reduction and carbon benefits are unverified

At a fixed 10% scenario threshold: 6 missed events and 9 false alarms.
Low probability still carries risk.

A 90-day validation plan

Pending plan by two students. Recruitment has not started.

Weeks 1 to 2	Weeks 3 to 4	Weeks 5 to 8	Weeks 9 to 12
Needs check →	**Understanding** →	**Shadow run** →	**Review**
Recruit 5 to 8 willing participants	Fixed historical cases Log misconceptions	Check forecast access Log failures and gaps	Review tasks and failures Continue, revise or stop

Planned target: at least 80% complete the task within 3 minutes and explain the key limits correctly.

Small-sample usability study. This 90-day period does not cover the next full Meiyu season.

Meiyu Prepare

Next: test whether merchants can complete and understand the comparison

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English demo

kongbaigit.github.io/meiyu-ai4climate/en/

English video

kongbaigit.github.io/meiyu-ai4climate/en/promo/

Code and evidence

github.com/kongbaigit/meiyu-ai4climate